Ryerson University

Department of Mathematics

Sample Test # 3

MTH 240 – Calculus II

Student Number: _____. Last Name:_____. First Name (**Print**):_____.

Signature:_____.

Professor (circle one)

Dr Bonato	Dr Ha
Dr Olivares	Dr Tasić

Date: April 17, 2009, 11:30 am

Duration: 1hr. 30 min.

Section : _____

Instructions:

1. Have your student card available on your desk.
2. This is a closed-book test. **Notes, calculators and other aids are not permitted.** Verify that your test booklet has pages 1-6.
3. Do not separate the pages of this test booklet.
4. The point value of each question is indicated by the question number.
5. Include all significant steps in your solutions to the questions, presented in the correct order. **Unjustified answers will be given little or no credit.** Cross out or erase all rough work not relevant to your solution.
6. Present your solutions neatly and legibly in the space provided. **Messy or illegible solutions will receive no credit.**
7. If you need more space, use the back of the previous page. Indicate this fact on the original page.

For Instructor's use only.

Question	Mark
1	
2	
3	
4	
5	
Total	/50

1. [10 marks] Find the radius of convergence and interval of convergence of the series

$$\sum_{n=0}^{\infty} \frac{(x-2)^n}{n^2+1}.$$

If $a_n = \dfrac{(x-2)^n}{n^2+1}$, then $\lim_{n\to\infty} \left| \dfrac{a_{n+1}}{a_n} \right| = \lim_{n\to\infty} \left| \dfrac{(x-2)^{n+1}}{(n+1)^2+1} \cdot \dfrac{n^2+1}{(x-2)^n} \right| =$

$= |x-2| \lim_{n\to\infty} \dfrac{n^2+1}{n^2+2n+2} = |x-2|$. By the Ratio Test, the series $\sum_{n=0}^{\infty} \dfrac{(x-2)^n}{n^2+1}$ converges when $|x-2| < 1$ [Radius $R = 1$]

$|x-2| < 1 \Leftrightarrow -1 < x-2 < 1 \Leftrightarrow 1 < x < 3$.

when $x = 1$, the series $\sum_{n=0}^{\infty} (-1)^n \dfrac{1}{n^2+1}$ converges by the AST

$b_n = \dfrac{1}{n^2+1}$, $\lim_{n\to\infty} b_n = 0$ and $b_{n+1} \leq b_n$

when $x = 3$, the series $\sum_{n=0}^{\infty} \dfrac{1}{n^2+1}$ converges by comparison with p-series $\sum_{n=0}^{\infty} \dfrac{1}{n^2}$ [$p = 2 > 1$].

Hence, the interval of convergence is $I = [1, 3]$

2. [(6+4) marks]

(a) Find a power series representation for $f(x) = \arctan \frac{1+x}{1-x}$ and determine the interval of convergence.

$$f(x) = \arctan \frac{1+x}{1-x} \Rightarrow f'(x) = \frac{1}{1+\left(\frac{1+x}{1-x}\right)^2} \cdot \left(\frac{1+x}{1-x}\right)' = \frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = \sum_{n=0}^{\infty} (-x^2)^n$$

$$= \sum_{n=0}^{\infty} (-1)^n x^{2n} \quad \text{for } |-x^2| = |x^2| < 1 \Leftrightarrow |x| < 1.$$

Now $f(x) = \int \left(\sum_{n=0}^{\infty} (-1)^n x^{2n} \right) dx = \sum_{n=0}^{\infty} (-1)^n \int x^{2n} dx = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} + C$

with $R = 1$ and $|x| < 1$.

Since $f(0) = \arctan \frac{1+0}{1-0} = \arctan 1 = \frac{\pi}{4} \Rightarrow f(0) = C = \frac{\pi}{4}$.

If $x = -1$ $f(x) = \frac{\pi}{4} + \sum_{n=0}^{\infty} (-1)^n \frac{(-1)^{2n+1}}{2n+1} = \frac{\pi}{4} - \sum_{n=0}^{\infty} (-1)^n \frac{1}{2n+1}$ which converges by the AST

$b_n = \frac{1}{2n+1}$, $\lim_{n\to\infty} b_n = 0$, $b_{n+1} \leq b_n$.

So, $f(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} + \frac{\pi}{4}$ with $R = 1$ and $I = [-1, 1)$

(b) Find the Taylor polynomial $T_3(x)$ for $f(x) = x \ln x$ at $a = 1$.

n	$f^{(n)}(x)$	$f^{(n)}(1)$
0	$x \ln x$	0
1	$\ln x + 1$	1
2	$1/x$	1
3	$-1/x^2$	-1

$$T_3(x) = (x-1) + \frac{1}{2}(x-1)^2 - \frac{1}{6}(x-1)^3$$

3. [(5+5) marks] Let $z = f(u) + g(v)$ and $u = x + at$, $v = x - at$, where f and g are assumed to be at least twice differentiable and a is any constant.

(a) Find $\frac{\partial z}{\partial t}$ and $\frac{\partial z}{\partial x}$.

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial t} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial t} = f'(u) \cdot a + g'(v) \cdot (-a) = a f'(u) - a g'(v)$$

$$\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x} = f'(u) + g'(v)$$

(b) Find $\frac{\partial^2 z}{\partial t^2}$ and $\frac{\partial^2 z}{\partial x^2}$ and show that z satisfies

$$\frac{\partial^2 z}{\partial t^2} = a^2 \frac{\partial^2 z}{\partial x^2}.$$

$$\frac{\partial^2 z}{\partial t^2} = \frac{\partial}{\partial t}\left(\frac{\partial z}{\partial t}\right) = \frac{\partial}{\partial u}\left(\frac{\partial z}{\partial t}\right) \cdot \frac{\partial u}{\partial t} + \frac{\partial}{\partial v}\left(\frac{\partial z}{\partial t}\right) \frac{\partial v}{\partial t} = a f''(u) \cdot a + (-a g''(v)) \cdot (-a)$$

$$= a^2 f''(u) + a^2 g''(v)$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial}{\partial u}\left(\frac{\partial z}{\partial x}\right) \cdot \frac{\partial u}{\partial x} + \frac{\partial}{\partial v}\left(\frac{\partial z}{\partial x}\right) \cdot \frac{\partial v}{\partial x} = f''(u) \cdot 1 + g''(v) \cdot 1$$

$$= f''(u) + g''(v)$$

So, $$\frac{\partial^2 z}{\partial t^2} = a^2 (f''(u) + g''(v)) = a^2 \frac{\partial^2 z}{\partial x^2}$$

4. [(5+5) marks]

If $u = x^4y + y^2z^3$, where

$$x = rse^t,\ y = rs^2e^{-t},\ z = r^2s\sin t$$

find the value of $\frac{\partial u}{\partial s}$ when $r = 2$, $s = 1$, $t = 0$.

$$\frac{\partial u}{\partial s} = \frac{\partial u}{\partial x}\cdot\frac{\partial x}{\partial s} + \frac{\partial u}{\partial y}\cdot\frac{\partial y}{\partial s} + \frac{\partial u}{\partial z}\cdot\frac{\partial z}{\partial s}$$

$$= (4x^3y)\,re^t + (x^4+2yz^3)\,2rse^{-t} + 3y^2z^2(r^2\sin t)$$

when $r=2$, $s=1$ and $t=0$ we have $x=2, y=2$ and $z=0$.

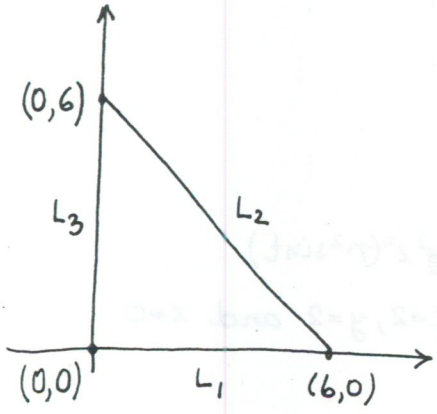
So, $\frac{\partial u}{\partial s} = 64\cdot 2 + 16\cdot 4 + 0\cdot 0 = 192$

(a) Find the partial derivative $\frac{\partial z}{\partial x}$ of the implicitly defined function $yz = \ln(x+z)$.

Let $F(x,y,z) = yz - \ln(x+z) = 0$.

Then $\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{-\frac{1}{x+z}}{y - \frac{1}{x+z}} = \frac{1}{y(x+z)-1}$

5. [10 marks] Find the absolute maximum and minimum values of $f(x,y) = 4xy^2 - x^2y^2 - xy^3$ on the closed triangular region in the xy-plane with vertices $(0,0)$, $(0,6)$, $(6,0)$.



We first solve inside the region.

Here $\frac{\partial f}{\partial x} = 4y^2 - 2xy^2 - y^3$, $\frac{\partial f}{\partial y} = 8xy - 2x^2y - 3xy^2$

$\frac{\partial f}{\partial x} = 0 \Rightarrow y^2(4-2x-y) = 0 \Rightarrow y = 0$ OR $4-2x = y$.

But $y=0$ isn't inside.

Substitute $y = 4-2x$ into $\frac{\partial f}{\partial y} = 0 \Leftrightarrow xy(8-2x-3y) = 0$

$\Rightarrow x = 0$ OR $4-2x = 0$ OR $8-2x-12+6x = 0$

$x = 0$ OR $x = 2$ OR $x = 1$

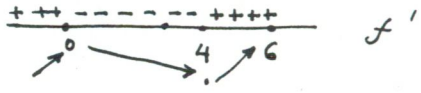
$x=0$ isn't inside, when $x=2$ we get $y=0$ and $(2,0)$ isn't inside

Thus the only critical point inside is $(1,2)$ and $f(1,2) = 4$.

On L_1, $y=0$ $f(x,0) = 0$.

On L_2, $x = -y+6$ and $f(-y+6, y) = -2y^2(6-y) = -12y^2 + 2y^3$

$f' = -24y + 6y^2 = 6y(y-4)$



Critical points are $y=0$, $y=4$

$f(6,0) = 0$ and $f(2,4) = -64$

On L_3, $x=0$, $f(0,y) = 0$.

So, on the whole region f attains the absolute max at $(1,2)$ with $f(1,2) = 4$ and the absolute min at $(2,4)$ with $f(2,4) = -64$.